Learning Beyond the IID Closed-Set Setting

Rana Taqveem ul Hassan (25280030)

Abstract

Real-world images are full of variance and interventions. For any image processing model these variations pose a hard challenge to solve as their performance deteriorates drastically under visual intervention, domain shifts and unseen classes. This report investigates such variations across four connected studies. First, it evaluated ResNet, ViT and CLIP models on STL-10 under various interventions like colour changes, translations, patch shuffling and cue conflicts. Model performance remained stable under colour and translation changes, however, disrupting spatial structure of images reduces their accuracy by 6.2–13.8 percentage points. On cue conflicts, strong shape bias has been observed across all models, while texture also had an influence but not strong enough to change the final decisions. Second, unsupervised domain adaptation is analysed on the PACS dataset, with Sketch as the target domain. A 38.61 point macro-F1 domain gap is observed on source-only ERM. DAN and CDAN improved target recognition; however, class-specific negative transfer is also visible, whereas DANN suffered from optimisation instability. Third, it examined target-free domain generalisation on sketches using DAN-DG and SAM. DAN-DG exhibited the best Sketch macro-F1 of 72.37% exceeding the target-aware DAN, however with stronger alignment push collapsed class-discriminative structure. SAM, in contrast, produced the lowest local sharpness proxy but failed to achieve better target performance, showing that local flatness of loss alone cannot achieve domain generalisation. Finally, the study examined open-set recognition on CIFAR-10. It demonstrated that semantically near unknowns are harder to reject than far ones. Among MSP, MLS, Energy and Mahalanobis, Mahalanobis distance performed the best on far unknowns. GCSC provided modest overall generalisation whereas PROSER offered limited meaningful rejections, given the experiment settings. In a nutshell, the observations show that domain invariance, confidence and local stability are not sufficient individually for reliable generalisation, as generalisation also depends on preserving the core semantic class structure, stable optimisation and the nature of the domain shift. Code: GitHub repository

Introduction and Research Questions

Modern vision models show strong performance under closed-set evaluation; however, they tend to struggle with real-world inputs because of the inherent complexity of real-world data. This report examines how different model architectures and experimental settings affect performance when input images differ in appearance, domain, or class composition. It evaluates robustness to visual interventions, domain adaptation, domain generalization, and unknown-class detection. It also investigates which features models rely on and whether alignment and local stability improve predictions on unseen domains and classes.

1 Task 1: Inductive Biases and Feature Representations

1.1 Experimental Setup

ResNet-50 [He et al., 2016], ViT-B/16 [Dosovitskiy et al., 2021] and CLIP [Radford et al., 2021] heads and zero-shot CLIP are tested on 500 STL-10 images [Coates et al., 2011] under colour, translation, patch-shuffle and AdaIN [Huang and Belongie, 2017] cue-conflict interventions [Geirhos et al., 2019] (Appendix Table 11).

1.2 Model's Reliance on Shape, Texture, and Color

Table 1: Accuracy (Acc, %), change from clean (Δ, pp) and prediction consistency (C, %) on 500 STL-10 test images. †: significant (paired McNemar, Holm, $p < 0.05$).

	Clean	**Grayscale**			**Hue** 30°			**Patch shuffle**		
Method	Acc	Acc	Δ	C	Acc	Δ	C	Acc	Δ	C
ResNet-50 head	97.8	96.4	−1.4	97.0	96.2	−1.6	97.0	90.0	−7.8†	91.8
ViT-B/16 head	98.8	96.6	−2.2	96.6	98.0	−0.8	98.4	92.6	−6.2†	92.8
CLIP ViT-B/32 head	97.8	95.4	−2.4	95.6	97.8	+0.0	97.4	84.0	−13.8†	83.6
CLIP ViT-B/32 zero-shot	96.8	94.4	−2.4†	96.8	95.8	−1.0	98.4	84.0	−12.8†	84.4

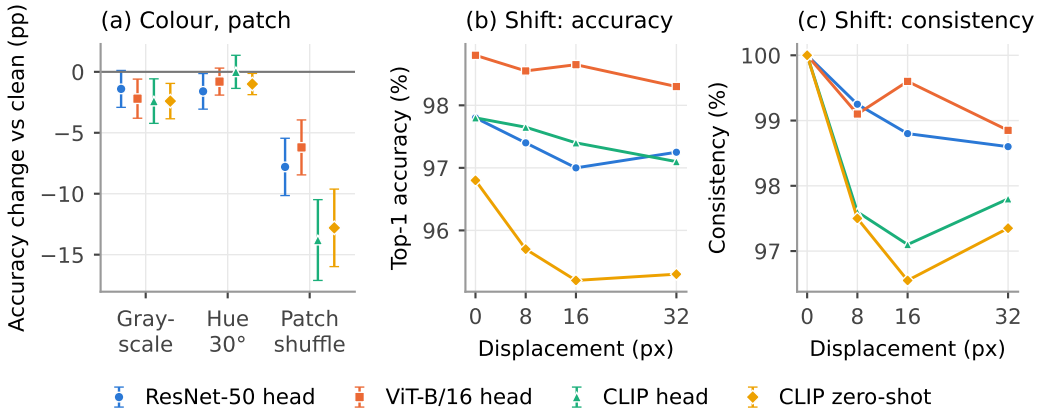


Figure 1: (a) Accuracy change from clean with 95% CIs. (b) Accuracy and (c) consistency against translation, averaged over four directions.

Model performance under interventions like color changes (greyscaling or hue modifications) showed no substantial drop in overall performance, suggesting that none of the models gave much importance to the color when making predictions. Removing color altogether cost between 1.4 and 2.4 points, and changing color impacted even less, from 0 to 1.6 points. Although, under the greyscale all four models broke more predictions than they managed to fix, only the zero-shot CLIP change was statistically more reliable.

When the texture of one class is pushed over the shape of another every model showed a strong inclination towards shape bias, with ResNet being the least biased towards shape and appearing to give some importance to texture as well as compared to the rest. Coverage of each model on the cue conflicts gives a different base for these percentages. For ResNet the shape bias is computed from 160 predictions, while for the others from about 180. A lower coverage means the model committed to a cue on fewer images. The models, ViT, the CLIP head, and zero-shot CLIP, predicted a third class on only 17 to 20 of the 200 conflicts, while ResNet did so on 40. Thus, ResNet's shape bias is measured less precisely than the others (Table 2). However, it cannot be claimed that ResNet's lower coverage follows from its greater reliance on texture or from some other weakness, since a third-class prediction only shows that neither cue won, not which cue caused it.

For the three trained heads (ResNet +14.5, CLIP head +12.0, ViT +7.0 points) the texture class rose into the top two of the ranked class scores more often under conflict than on the clean content image, so texture does influence their scores even though shape still dominates the final decision; zero-shot CLIP showed no such reliable movement.

Table 2: Cue-conflict decisions on 200 AdaIN images. Shape bias $= N_s/(N_s + N_t)$, coverage $= (N_s + N_t)/200$; 95% intervals in brackets.

Method	Shape	Texture	Other	Decisions	Shape bias (%)	Coverage (%)
ResNet-50 head	139	21	40	160	86.9 [81.4, 91.9]	80.0 [73.9, 85.0]
ViT-B/16 head	179	4	17	183	97.8 [95.6, 99.5]	91.5 [86.8, 94.6]
CLIP head	171	11	18	182	94.0 [90.2, 97.2]	91.0 [86.2, 94.2]
CLIP zero-shot	172	8	20	180	95.6 [92.2, 98.3]	90.0 [85.1, 93.4]

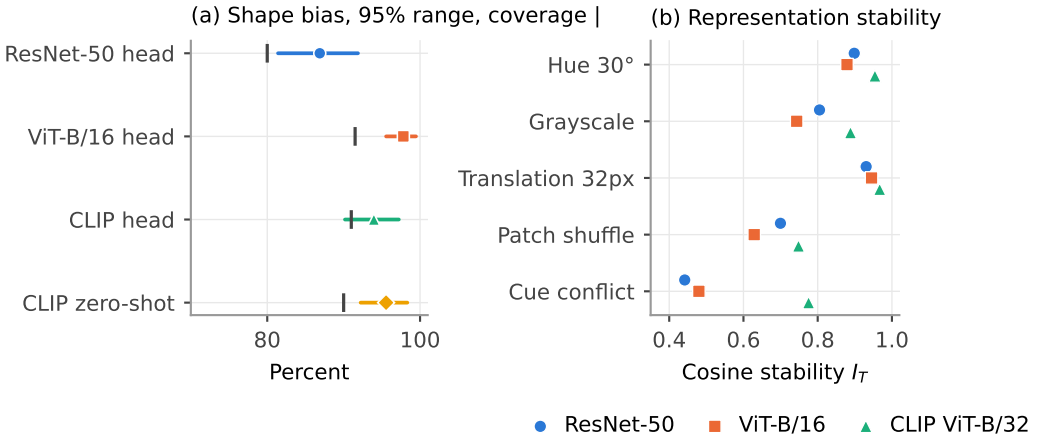


Figure 2: (a) Shape bias with 95% interval and coverage (|). (b) Cosine stability between clean and transformed features (1 = unchanged).

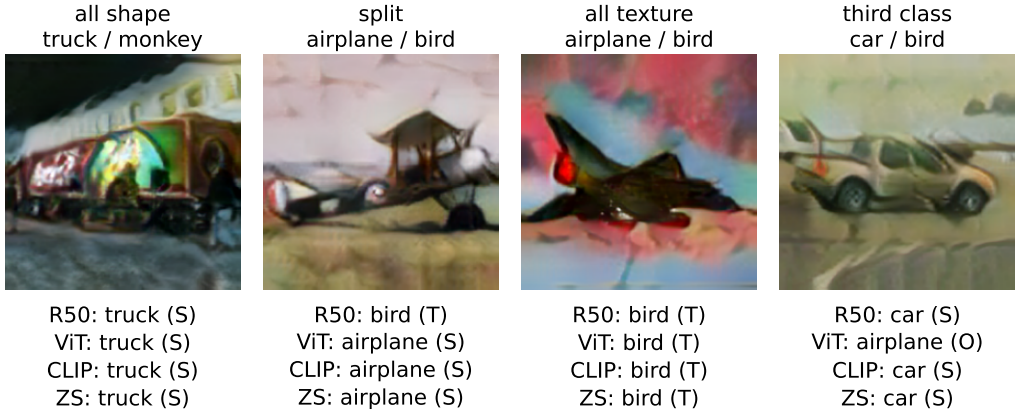


Figure 3: One seeded example per decision pattern (shape / texture class) with each method's prediction. S: shape, T: texture, O: other; ZS: zero-shot CLIP.

1.3 Spatial and Positional Invariance

Models show almost no drop in accuracy and performance with translations. Out of the 48 translation tests (8, 16, and 32 pixels across four directions and four methods), only 8 resulted in 10 or more predictions flipping to negative, with a maximum of 16 (zero-shot CLIP at 16 pixels left), and 7 of these 8 cases belong to zero-shot CLIP. The confidence intervals permit a performance decline of up to approximately 4 points, thus ruling out major degradation from translation while leaving smaller losses possible.

All four drops were statistically reliable, with ResNet and ViT models' performance declining by 6.2 to 7.8 points, where ResNet failed on 39 correct baseline predictions and was unable to correct a single incorrect baseline prediction. The CLIP models appeared to be hit the worst, with the CLIP head breaking 75 predictions while fixing only 6, and overall performance decreasing by 12.8 to 13.8 points. This difference is reliable, as both CLIP methods lost significantly more than ResNet and ViT.

The variation in model performance can be attributed to the changes introduced by each intervention. During the translation interventions, most of the global image and shape structure is preserved. Models still have access to enough global representation, evident from the feature similarity between 0.93 and 0.98 when comparing baseline and translated images, and they therefore retain their accuracy and consistency. This changes substantially with patch shuffling, which moves the patches around and destroys the global spatial arrangement: representational similarity falls to between 0.63 and 0.75, and all models lose accuracy. The correspondence holds when comparing the two interventions, but not when comparing models, since CLIP keeps the highest similarity under shuffling (0.748) while losing the most accuracy (-13.8 points), and ViT has the lowest similarity (0.629) with the smallest loss (-6.2 points). Furthermore, local features retain significant diagnostic value independently: since patch shuffling keeps every pixel and local texture within each block intact, the models maintain an accuracy of 84.0% to 92.6% despite the loss of spatial layout—substantially above chance level. This demonstrates that local texture information sustains the majority of the classification performance, whereas global structure provides the final 6.2 to 13.8 percentage points.

1.4 Representation Space Analysis

For all models, feature representation similarity remained quite stable across translations. For color interventions it drops marginally. The clearest divergence appears with the cue conflicts, where ResNet's representation moves the furthest of all (dropping to 0.442), while its predictions change only slightly, since 139 of 200 still follow the shape class. A possible explanation is that cosine similarity measures how far the representation moved, not what it still contains, so a large movement does not necessarily remove the information the classifier relies on. In these cases, prediction stability and representation stability come apart.

Another interesting finding is observed for ViT on the cue conflicts, where the similarity is 0.505 for stable predictions against 0.264 for flipped ones, showing that the images whose predictions flipped had moved further than those that held. However, under patch shuffling that gap nearly vanishes (0.014 for ResNet), so feature movement barely explains which predictions broke there. Representation change and prediction change therefore agree within a condition, while diverging across conditions.

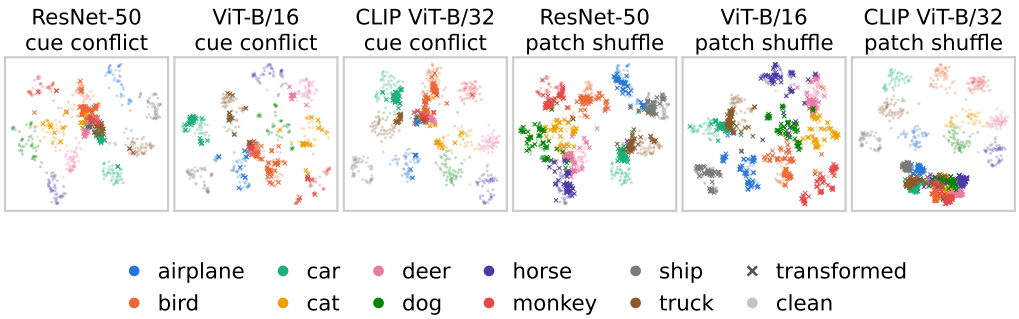


Figure 4: Joint t-SNE [van der Maaten and Hinton, 2008] of clean (dots) and transformed (crosses) features per backbone, coloured by class; coordinates are comparable only within a backbone.

CLIP vs CLIP Zero-Shot predicted same class for the same image and they share similar accuracy on the baseline, translations and colour changes. The visible differences come from patch shuffle and cue conflicts. Under patch shuffle the accuracy numbers of the two are identical (84.0%), yet their agreement on the predicted class drops to 88.2%, meaning they disagreed on 59 of 500 images, with 22 correct only for the head and 22 correct only for zero-shot. Under cue conflicts agreement is similarly low at 88.0%, and the texture-rank analysis shows that the transferred texture reliably raised the texture class in the head's top two (+12.0 points) but not in zero-shot's (+0.5, not reliable). With same backbone embeddings, the two decision rules read those same representations differently, and the trained head is the more texture-sensitive of the two.

1.5 Architecture and Observation Relationships

The models differ across their architecture, pretraining data, supervision, and capacity. Such cross-model differences are therefore confounded and cannot be attributed to architecture alone.

Across this experimentation, the one causal claim that can be made about the decision rule comes from comparing CLIP with a trained linear head against zero-shot CLIP. Since both share the same frozen backbone, the same pretraining and identical image embeddings, the observed differences must be driven by the decision rule rather than the representation, as highlighted in their comparison earlier.

A partial architectural link can be drawn between ResNet and ViT. Both were pretrained on the same dataset (ImageNet-1k) with the same supervision (class labels), so data and supervision are held constant. ResNet's reliably lower shape bias (86.9% vs 97.8%) and lower coverage (80.0% vs 91.5%, driven by 40 third-class predictions concentrated in cases like monkey-to-truck and deer-to-car) indicate a relatively greater reliance on texture than the other three, which is consistent with the convolutional inductive bias reported in the literature. However, the two models also differ in capacity (25M vs 87M parameters) and training recipe (the `IMAGENET1K_V2` ResNet weights use heavier augmentation), so architecture cannot be isolated as the cause. The shape bias built into our conflict set raises the absolute level for every method equally, so it explains the high values overall but not the difference between models.

Conversely, CLIP's substantially larger accuracy degradation under patch shuffle (-12.8 to -13.8 points, compared with -6.2 to -7.8 for ViT and ResNet) cannot be assigned to architecture, since CLIP differs from ViT in patch size (32 px vs 16 px), supervision (multimodal contrastive) and pretraining scale (400M web pairs) at the same time. The fact that both CLIP variants suffer equally does show that the effect comes from the representation rather than the decision rule.

2 Task 2: Unsupervised Domain Adaptation

2.1 Experimental Setup

A ResNet-18 is adapted from PACS [Li et al., 2017] sources to Sketch with DAN [Long et al., 2015] (MMD [Gretton et al., 2012]), DANN [Ganin et al., 2016] and CDAN [Long et al., 2018], selected on source validation only (Appendix Table 12).

Table 3: Source validation macro-F1, Sketch results and source-vs-target separability (Sep., chance 50%). Lower rows: DANN $\alpha_{\max}$ study (plot: Appendix Fig. 11).

	Source validation (macro-F1)				**Target: Sketch**			
Method	Photo	Art	Cartoon	Mean	Acc	Macro-F1	ΔAcc	**Sep.**
Source-only ERM	95.53	92.64	95.37	94.51	55.71	55.90	—	99.86
DAN (MMD)	94.88	88.85	94.56	92.76	67.80	**66.84**	+12.09	**89.70**
DANN ($\alpha_{\max} = 1$)	86.73	80.30	83.25	83.43	41.51	36.71	−14.20	100.00
CDAN	93.87	90.26	94.12	92.75	62.36	63.12	+6.64	98.49
Controlled study: maximum gradient-reversal strength in DANN								
DANN ($\alpha_{\max} = 0.25$)	96.58	91.31	94.61	94.17	28.23	20.40	−27.49	100.00
DANN ($\alpha_{\max} = 0.5$)	94.42	87.15	91.10	90.89	38.92	32.60	−16.80	99.86

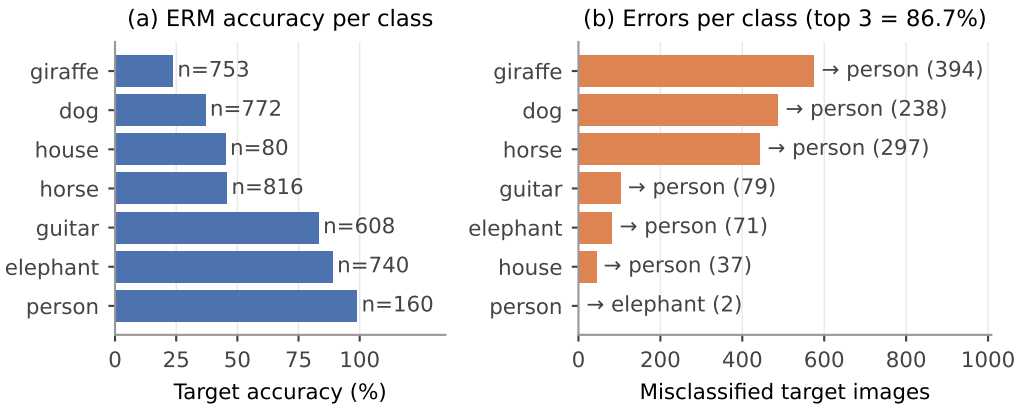


Figure 5: Source-only ERM on Sketch: (a) per-class accuracy with support n; (b) misclassified images per class and the class that absorbs most of them.

2.2 The Source-Only Domain Gap and Its Failure Modes

The source-only ERM showed a substantial domain gap of 38.61 points (macro-F1) between the source and target domains. Table 3 demonstrates that ERM made the model learn class-discriminative source-domain features; however, these features are not directly transferable to the target domain and contain strong domain-specific information, which is clearly visible in the 99.86% domain-separability score.

The most important target failures were found by accounting for the overall error share of each class: Giraffe, Dog, and Horse accounted for 86.7% of the overall errors on the target domain. Additionally, the model also displayed a strong bias towards the person class, and the model confused the majority of the classes with person, producing around 1,116 such errors. In contrast, the elephant and guitar classes showed good results, obtaining 88.78% and 83.06% accuracy. This shows that the domain shift did not affect each class uniformly.

2.3 Domain Separability and Target Recognition

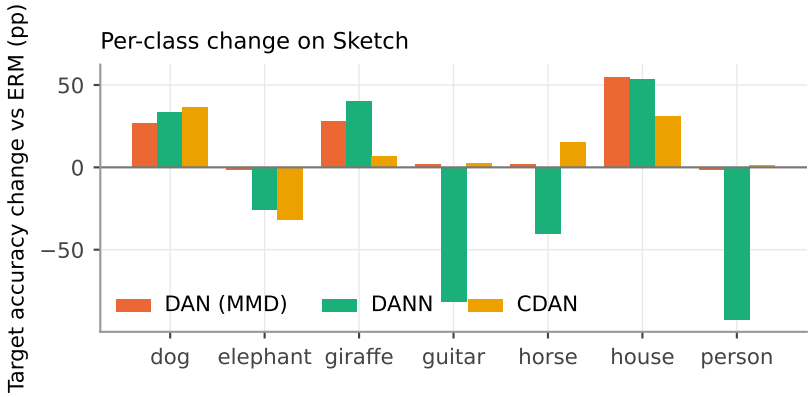


Figure 6: Per-class Sketch accuracy change relative to ERM (separability plot: Appendix Fig. 10).

Across all methods, we see a clear enough pattern: the higher the domain separability between source and target domains, the lower the target accuracy scores. The treatment methods like DAN and CDAN were able to fractionally reduce this separability gap, and as the domain separability scores reduced with these methods, so did the accuracy on the target domain improve. DANN, though it performed miserably because of an unstable optimization flow, shows that it has the highest domain separability score and thus has the lowest target-domain accuracy.

DAN or CDAN showed better adaptation than others. However, this improvement was not uniform across all of the classes. In DAN, the adaptation improved 5 out of 7 classes, with giraffe improving by 27.89 points, house by 55.00 points, and dog by 26.81 points. However, for elephants and persons, it exhibited negative transfer, declining in performance by 1.62 and 1.25 points. Similarly,

CDAN also adapted successfully overall and improved 6 out of 7 classes, except elephant, where its performance decreased by 31.62 points. DANN displayed negative transfer overall and reported terrible performance on guitar, person, horse, and elephant. Its target macro-F1 plummeted by 19.19 points relative to ERM, most likely because of the instability it faced during optimization.

Overall, reduced domain separability can be associated with better performance on target recognition for DAN and CDAN.

2.4 Class-Conditional versus Marginal Alignment

The results do not clearly support the claim that class conditioning helps preserve class semantic features better. With DAN, the domain separability score reduced down and showed the best target adaptation performance, however, the individual class-wise macro-F1 for DAN is not uniform across the classes.

CDAN also observed an improved domain adaptation and reached a 63.12% target macro-F1, marginally less than DAN. Overall, it performed well on 6 out of 7 classes and also preserved person and guitar recognition, improving them by 1.25 and 2.47 points, but reduced the adaptation on the elephant class by a massive 31.62 points, and also provided just a 6.51-point gain on the giraffe, whereas on the same giraffe class DAN improved the adaptation by 27.89 points. Thus, the class conditionality might have partially preserved the class-wise consistency, with 6 classes improving; however, it did not manage to improve the overall aggregate for target adaptation.

CDAN does outperform the marginal adversarial alignment DANN, where DANN only reported 36.71% macro-F1. Overall results may suggest that conditioning can better preserve semantic structure, and it improved aggregate target adaptation relative to ERM but did not outperform DAN, and stable DAN depicted a good marginal alignment that may outperform CDAN.

2.5 Effect of Increasing Alignment Strength

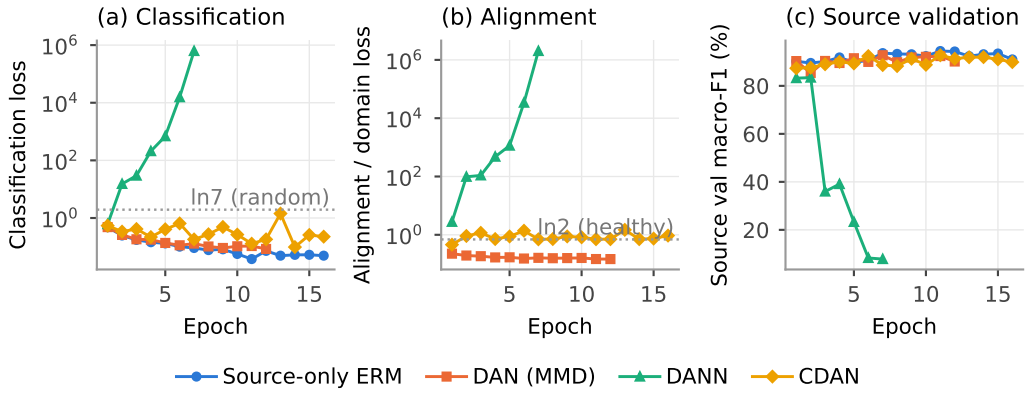


Figure 7: (a) Classification loss (dotted: $\ln 7$). (b) Alignment loss (dotted: $\ln 2$). (c) Mean source-validation macro-F1.

With DANN(1.0) (Figure 7) from epoch 2 onwards, the overall optimization became unstable, as the model's classification loss shot up to 15.21 in the second epoch; the training accuracy dropped to 69.80%, while the validation F1 moved up to 83.43%, which tells us the model is making some very confident errors as its training accuracy decreases while its validation score remains high. Source-validation performance then collapsed outright from the 3rd epoch, when the validation macro-F1 plummeted from 83.43% down to 35.96% and then eventually reached 7.83%. The domain accuracy stayed between 42.29% and 58.18%; however, the alignment loss increased from 2.78 at the end of the first epoch to 97.49 in the second, reaching 2,013,773.19 in 7 epochs.

As we halve DANN's alignment strength, things do start to get a little better for the source-domain accuracy, as the source macro-F1 gained 7.46 points; however, the reduced alignment strength impacted the target macro-F1, decreasing it by 4.11 points. The pattern repeats when we further reduce the alignment strength, which further increases the model accuracy on source classification; however, it progressively damages target recognition, decreasing target macro-F1 by another 12.19 points. Interestingly, while we have these variations in target F1 score as we decrease the alignment

strength, source and target separability remains around 100%. Therefore, in this scenario, the stronger or weaker adversarial objective failed to produce any measurable domain-invariant features.

In fact, all three DANN configurations remained well below the ERM target macro-F1 of 55.90%. So, the trade-off between low and high alignment strength had a substantial positive effect on target macro-F1 within the DANN variants; however, stronger alignment destabilized the model's training, and a smaller value further deteriorated the target macro-F1.

3 Task 3: Domain Generalization

3.1 Experimental Setup

The Task 2 protocol is reused without Sketch until final evaluation [Gulrajani and Lopez-Paz, 2021], comparing ERM, pairwise-source DAN-DG and SAM [Foret et al., 2021] (Appendix Table 13).

Table 4: Source validation accuracy / macro-F1 (%) per domain, mean and worst; λ_{DG} study below the rule.

Method	Photo	Art	Cartoon	Mean	Worst
ERM	96.11 / 95.53	92.68 / 92.64	94.67 / 95.37	94.49 / 94.51	92.68 / 92.64
DAN-DG ($\lambda_{\mathrm{DG}} = 1$)	94.31 / 93.24	89.02 / 89.35	92.11 / 92.09	91.82 / 91.56	89.02 / 89.35
SAM ($\rho = 0.05$)	97.31 / 96.77	92.44 / 92.34	95.74 / 96.23	95.16 / 95.12	92.44 / 92.34
Controlled study: alignment strength λ_{DG} in DAN-DG					
DAN-DG ($\lambda_{\mathrm{DG}} = 0.1$)	95.51 / 94.67	91.46 / 91.24	91.68 / 92.17	92.89 / 92.69	91.46 / 91.24
DAN-DG ($\lambda_{\mathrm{DG}} = 10$)	41.32 / 21.54	25.12 / 15.39	18.12 / 11.24	28.19 / 16.06	18.12 / 11.24

Table 5: Sketch results, change against ERM (pp), source separability (chance 33.3%) and the sharpness proxy: loss $\mathcal{L}$, loss after a radius-0.05 ascent step $\mathcal{L}_{+\epsilon}$, and Δ_{sharp} (96 images). Plots: Appendix Figs. 13 and 14.

	Sketch					**Sharpness proxy**		
Method	Acc	Macro-F1	ΔAcc	ΔF1	**Sep.**	$\mathcal{L}$	$\mathcal{L}_{+\epsilon}$	Δ_{sharp}
ERM	55.71	55.90	—	—	86.38	0.192	0.792	0.600
DAN-DG ($\lambda_{\mathrm{DG}} = 1$)	68.44	72.37	+12.73	+16.46	80.07	0.217	0.849	0.631
SAM ($\rho = 0.05$)	64.55	67.64	+8.83	+11.74	86.38	0.073	0.214	0.141
Controlled study: alignment strength λ_{DG} in DAN-DG								
DAN-DG ($\lambda_{\mathrm{DG}} = 0.1$)	71.70	72.63	+15.98	+16.73	81.40	0.150	0.887	0.737
DAN-DG ($\lambda_{\mathrm{DG}} = 10$)	10.56	7.84	−45.15	−48.06	48.84	1.855	4.103	2.248

3.2 Source Validation as a Predictor of Sketch Performance

Table 6: Per-class Sketch accuracy (%) with change against ERM (pp), including Task 2 DAN. Plots: Appendix Figs. 13 and 15.

			Task 3			**Task 2**
Class	Support	ERM	DAN-DG (λ=1)	SAM	DAN-DG (λ=10)	DAN
`dog`	772	36.79	36.66 (−0.13)	30.18 (−6.61)	11.79 (−25.00)	63.60 (+26.81)
`elephant`	740	88.78	96.22 (+7.43)	98.51 (+9.73)	30.41 (−58.38)	87.16 (−1.62)
`giraffe`	753	23.51	44.89 (+21.38)	56.31 (+32.80)	0.00 (−23.51)	51.39 (+27.89)
`guitar`	608	83.06	95.56 (+12.50)	85.36 (+2.30)	0.66 (−82.40)	84.87 (+1.81)
`horse`	816	45.59	73.04 (+27.45)	55.64 (+10.05)	0.00 (−45.59)	47.67 (+2.08)
`house`	80	45.00	93.75 (+48.75)	81.25 (+36.25)	63.75 (+18.75)	100.00 (+55.00)
`person`	160	98.75	65.00 (−33.75)	70.00 (−28.75)	27.50 (−71.25)	97.50 (−1.25)

Good performance on the source domain does not guarantee good performance on the target domain, and this is visible via the results (Tables 4 and 5). DAN-DG's and SAM's worst-performing source macro-F1 is 89.35 and 92.34, respectively. While establishing a possible link between the validation numbers of source domains and the test numbers of the target domains, ERM, as anticipated, performed worst in the target domain, whereas DAN-DG has the lowest mean source accuracy of 91.82 and best target accuracy/macro-F1 of 68.44/72.37. SAM has performed the highest mean

source accuracy of 95.16, but still its target accuracy is 3.89 points lower than DAN-DG. Therefore, it is difficult to establish that source validation performance reliably predicts performance on Sketch.

ERM performs very poorly on Sketch for giraffe, dog, horse, and house, with accuracies of 23.51%, 36.79%, 45.59%, and 45.00%, whereas in the case of DAN-DG, 5 out of 7 classes gained points (house by 48.75 points, horse by 27.45, giraffe by 21.38, and guitar by 12.50). However, for the person, the performance declined by 33.75 points. Therefore, high aggregate source performance hides substantial target-specific class failures, particularly the degradation of person and the continued difficulty of dog recognition.

The art domain is recorded as the worst across all three methods.

3.3 Source-Domain Invariance and Class-Discriminative Information

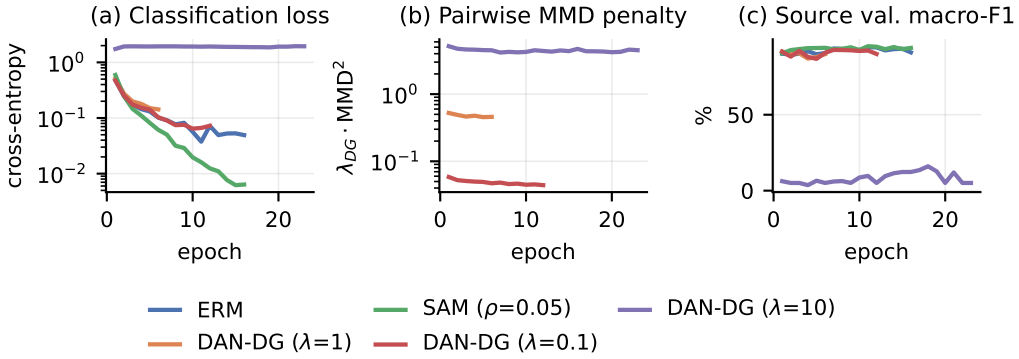


Figure 8: (a) Classification loss, (b) weighted pairwise MMD penalty, (c) mean source-validation macro-F1; ERM is the reused Task 2 run.

DAN-DG made the observed source domains slightly less distinguishable, as shown by the source-domain separability score, which decreases to 80.07% compared with 86.38% under ERM. Its performance on Sketch also got better, and it reported an increase in macro-F1 from 55.90 to 72.37. These numbers appear to support the claim; however, the results change drastically when λ_{DG} is increased to 10. The separability score fell to 48.84%, but the macro-F1 on both the Sketch and the source just collapsed to 7.84% and 16.06%, respectively. Class-wise, 6 out of 7 classes lost substantial points: guitar fell by 82.40 points, person by 71.25, elephant by 58.38, and both giraffe and horse reached 0% accuracy. Moreover, in the case of SAM and ERM, the separability is identical at 86.38%, yet SAM's Sketch macro-F1 is 67.64 against ERM's 55.90. This suggests that excessive source alignment does remove domain-discriminative features, making domains more difficult to distinguish; however, at the same time, it destroys the core semantic structure required for classification. Therefore, moderate invariance among the domains partially improves Sketch recognition, while a stronger push results in loss of critical class-discriminative information, resulting in poor performance on the target domain.

3.4 Local Stability and Sketch Performance

A lower common sharpness proxy under given perturbation partially supports the claim about local stability, but it does not establish that the overall loss surface is flatter and will generalize better on any domain shifts. With SAM perturbation the loss increased by just 0.141, as compared to ERM and DAN-DG, which reported a change of 0.600 and 0.631, respectively. Therefore, the local stability ranking is

$$\text{SAM} < \text{ERM} < \text{DAN-DG}.$$

However, it does not establish that the local stability ranking enables better performance on the target domain, because on the target domain the performance of each model is

$$\text{DAN-DG} > \text{SAM} > \text{ERM}.$$

DAN-DG reported the highest sensitivity to a minor perturbation; however, its Sketch macro-F1 is the highest (72.37%), whereas SAM is much flatter as per the proxy ranking, but it reaches 67.64%.

Nevertheless, SAM does improve over ERM by 8.83 accuracy points and 11.74 macro-F1 points, suggesting that local parameter stability may contribute to some extent to better generalization, but it is difficult to confirm that in every setting SAM will result in better performance on target domains.

3.5 Target-Aware DAN versus Target-Free DAN-DG

Both methods improve substantially over the shared ERM baseline (DAN: Table 3; ERM and DAN-DG: Table 5; per class: Table 6).

DAN used unlabeled Sketch data to align the source features and improved over ERM by 12.09 accuracy points and 10.93 macro-F1 points. DAN-DG, which never had access to the target data (sketches) during training, improved by 12.73 accuracy points and 16.46 macro-F1 points. DAN-DG exceeds target-aware DAN by 0.64 accuracy points and 5.53 macro-F1 points.

This suggests that access to unlabeled Sketch data did not provide a substantial empirical advantage. Instead, it highlights that DAN-DG was able to learn the cross-domain invariances, which resulted in source-target alignment. However, this does not establish that unlabeled Sketch data has no value, and several limitations prevent this from being a complete causal attribution.

The whole experiment is limited to a single seed, 6304, a different seed might have produced different results.

DAN and DAN-DG work on different distributions; the DAN objective aligns source and target domain features given by $\mathcal{L}_{\text{align}}^{\text{DAN}} = \text{MMD}^2\left(F(X_P \cup X_A \cup X_C), F(X_S)\right)$. whereas DAN-DG tries to learn invariances among the source domains:

$$\mathcal{L}_{\text{align}}^{\text{DAN}-\text{DG}} = \frac{\text{MMD}^2(P, A) + \text{MMD}^2(P, C) + \text{MMD}^2(A, C)}{3}.$$

In DAN, the pooled comparison between the source and target can hide the differences between the individual source domains; however, the pairwise terms in DAN-DG ensure that these differences are accounted for.

4 Task 4: Open-Set Recognition

4.1 Experimental Setup

A CIFAR-10 [Krizhevsky, 2009] ResNet-18 is scored with MSP [Hendrycks and Gimpel, 2017], MLS [Vaze et al., 2022], Energy [Liu et al., 2020] and Mahalanobis [Lee et al., 2018] against 16 CIFAR-100 unknowns; GCSC adds RandAugment [Cubuk et al., 2020] and PROSER [Zhou et al., 2021] uses mixup [Verma et al., 2019] placeholders (Appendix Table 14).

4.2 Semantic Similarity and Rejection

Table 7: Post-hoc scores on Vanilla: AUROC, and rejection at the threshold accepting 95% of CIFAR-10 validation images (Known: test images accepted; 800 images per unknown group).

	AUROC				Rejection (%)		
Score	Near	Far	All	Known (%)	Near	Far	All
MSP	**0.814**	0.887	0.850	94.09	31.38	46.38	38.88
MLS	0.802	0.883	0.842	94.22	**34.75**	**54.13**	**44.44**
Energy	0.802	0.883	0.843	94.41	34.50	53.25	43.88
Mahalanobis	0.795	**0.911**	**0.853**	94.36	27.75	51.38	39.56

The semantic similarity of certain classes made the rejection substantially harder for the model. With the vanilla MLS measure of unknownness, the model rejected only 34.75% near unknowns with an AUROC of 0.802, whereas it rejected around 54.13% far unknowns with an AUROC of 0.883.

The most frequently accepted unknown classes were among pickup trucks (91% accepted, most often as automobiles), bus (82%, most often as truck), and wolf (77%, most often as dog). Similarly, the model confused wolves as dogs and leopards as cats, indicating that the visual and semantic similarity

Table 8: Most confidently accepted unknowns (Vanilla, MLS; accepted below -6.058). Per-class acceptance: Appendix Fig. 16.

Near unknowns			Far unknowns		
Unknown	Predicted	Score	Unknown	Predicted	Score
wolf	dog	-12.12	keyboard	bird	-12.15
fox	cat	-12.04	mushroom	bird	-11.89
bus	truck	-11.60	chair	airplane	-11.77
tractor	ship	-11.54	keyboard	truck	-11.54
pickup truck	automobile	-11.46	sunflower	bird	-11.52

among some of the unknown categories with known ones is quite plausible, making rejection difficult for the model.

Among some of the semantically far unknowns, the wardrobe (68%), bowl (53%), and sunflower (47%) had the highest acceptance rates. Additional confident failures are the keyboard/mushroom by bird and chair by airplane. The high acceptance rates for these failed semantically far unknowns suggest that the classifier might have responded to incidental similarities in shape, color or some other spurious correlations like background rather than some discriminative class features.

4.3 What the Novelty Scores Capture

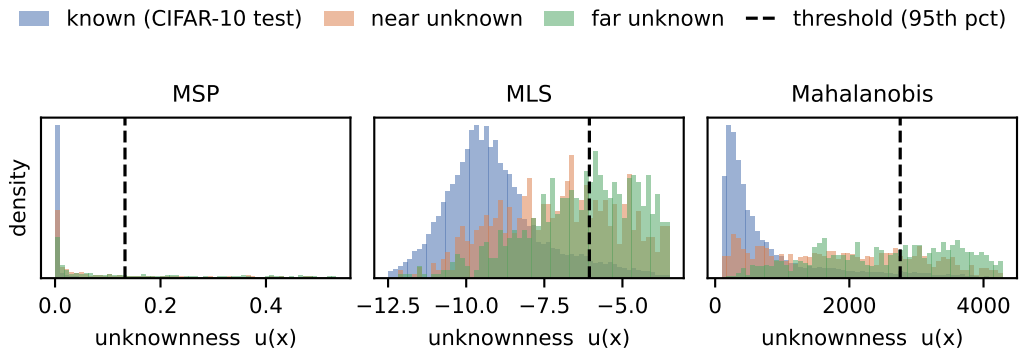


Figure 9: Unknownness distributions of MSP, MLS and Mahalanobis on Vanilla with the threshold (dashed); ROC curves: Appendix Fig. 17.

MSP measures relative softmax confidence, where the dominant probability is considered as known-like, and the probabilities spread across classes are unknowns. As compared to MSP, MLS measures the largest absolute logit. The maximum logit represents the strongest raw response of the classifier among the known classes. Energy uses all the classifier logits to compute unknownness. Mahalanobis measures distance from the nearest known-class feature cluster by calculating a shared diagonal covariance from the training features (Appendix Table 10).

Novelty scores for all four methods show that classifying far unknowns as unknown was much easier than near unknowns; MLS and energy methods produced almost identical results $0.802/0.883$ and $0.802/0.883$ near/far AUROC. This indicates that the largest logit dominated. MSP was most effective for near unknowns, with the highest AUROC of 0.814, which indicates that the relative probability has marginally more information for semantically similar unknowns.

Mahalanobis produced the best far AUROC (0.911) and the best overall AUROC (0.853) as well. It shows that for semantically far classes, the feature distance was particularly informative. However, on the contrary, it depicted the weakest near rejection because the near unknowns lie closer to the known clusters, as in the case of wolves and buses, which can lie close to dog or vehicle feature clusters.

Therefore, MSP softmax confidence was most useful for ranking near unknowns, feature distance was strongest for far unknowns, and using all logits through Energy added little beyond MLS in this experiment.

Table 9: Vanilla, GCSC and PROSER under MLS, plus PROSER's placeholder score; CSA uses the ten known-class logits.

		AUROC				Rejection (%)		
Model (score)	CSA (%)	Near	Far	All	Known (%)	Near	Far	All
Vanilla (MLS)	94.54	0.802	0.883	0.842	94.22	34.75	54.13	44.44
GCSC (MLS)	**95.12**	**0.826**	**0.909**	**0.868**	94.60	**37.13**	**59.00**	**48.06**
PROSER (MLS)	94.63	0.803	0.887	0.845	94.64	32.75	50.00	41.38
PROSER (placeholder)	94.63	0.780	0.886	0.833	94.91	29.88	48.25	39.06

4.4 Did GCSC Improve Closed-Set and Open-Set Performance?

GCSC shows a slight improvement on closed-set performance, where it gained 0.58 points over the vanilla model, which is about 58 images out of 10,000 test images, not a big number. Since we used only one seed for the dataset, this improvement is not something on which we can assert the claim that GCSC improves the closed-set performance. On open-set, the ordered ranking score AUROC on near and far shows a tiny improvement of 0.024 and 0.027, respectively.

While the AUROC gains were roughly the same for both near and far unknowns, at the rejection threshold for near and far unknowns model reports a gain of 2.38 points on near unknown rejections and 4.88 points on far unknown rejections. This suggests that the augmentation might have encouraged a kind of representation that is less dependent on superficial fluctuations in appearances and thus improved known-class recognition and detection of unknowns. However, the augmentation still does not fully resolve the semantic overlap between classes like bus and truck or wolf and dog, and near-semantic novelty remains the harder problem.

4.5 Did PROSER Improve Rejection, and What Was the CSA Trade-off?

The PROSER method via MLS reported an AUROC of 0.845, a 0.003 increase over Vanilla (0.842); however, it's still below GCSC (0.868) by 0.023. It reports near and far rejection rates as 32.75% and 50.00% respectively, on the lower side compared to both Vanilla $34.75\%/54.13\%$ and GCSC $37.13\%/59.00\%$. The closed-set accuracy of 94.63% is roughly the same as that of Vanilla (94.54%) and GCSC (95.12%).

PROSER's own placeholder scores were also weaker, especially on near unknowns compared to MLS, with an AUROC of 0.780 on near and 0.886 on far, and rejection rates of 29.88% and 48.25%, respectively. PROSER was fine-tuned for 50 epochs; however, the highest validation accuracy was achieved very early at the 4th epoch, which may explain why the placeholders were weaker, as at that moment the placeholders' loss was still relatively high; thus, the dummy classifiers had not yet learned stronger class boundaries. Later training did reduce the placeholders' loss; however, it destabilized the validation accuracy, which dropped as low as 53.12% at epoch 13 and ended at 93.40% at epoch 50, below the 95.20% best.

The models' reported results were not strong enough to clearly support the claim that the classifier and the data placeholders tightened the decision boundaries usefully. The interpolation between the known classes appears insufficient to represent the actual near and far unknown distributions. The model reported several confident failures, such as bus/truck and fox/cat, suggesting that the near unknown lies close to a single known class rather than between two knowns. On the other hand, the failures, such as the keyboard/bird and chair/airplane, show that far unknowns may also occupy directions not represented by interpolating known-class features.

5 Conclusion

Across the experiments, it is observed that a strong closed-set performance did not guarantee better generalization and robustness under the distribution shift or unknown classes. Successful generalization required preserving the semantic information. Moderate alignment and augmentation helped, while excessive alignment led to unstable optimization and caused negative transfer. Overall, domain invariance, confidence or local flatness were not able to predict reliably on the target domain, highlighting the need for multiple complementary experiments and evaluations beyond IID accuracy.

References

Adam Coates, Andrew Ng, and Honglak Lee. An analysis of single-layer networks in unsupervised feature learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011. URL https://proceedings.mlr.press/v15/coates11a.html.

Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. RandAugment: Practical automated data augmentation with a reduced search space. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020. URL https://arxiv.org/abs/1909.13719.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021. URL https://arxiv.org/abs/2010.11929.

Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations (ICLR)*, 2021. URL https://arxiv.org/abs/2010.01412.

Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016. URL https://arxiv.org/abs/1505.07818.

Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://arxiv.org/abs/1811.12231.

Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. URL https://jmlr.org/papers/v13/gretton12a.html.

Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations (ICLR)*, 2021. URL https://arxiv.org/abs/2007.01434.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. URL https://arxiv.org/abs/1512.03385.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017. URL https://arxiv.org/abs/1610.02136.

Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017. URL https://arxiv.org/abs/1703.06868.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. URL https://arxiv.org/abs/1807.03888.

Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy M. Hospedales. Deeper, broader and artier domain generalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017. URL https://arxiv.org/abs/1710.03077.

Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL https://arxiv.org/abs/2010.03759.

Mingsheng Long, Yue Cao, Jianmin Wang, and Michael I. Jordan. Learning transferable features with deep adaptation networks. In *International Conference on Machine Learning (ICML)*, 2015. URL https://arxiv.org/abs/1502.02791.

Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. URL https://arxiv.org/abs/1705.10667.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021. URL https://arxiv.org/abs/2103.00020.

Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008. URL https://jmlr.org/papers/v9/vandermaaten08a.html.

Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: a good closed-set classifier is all you need? In *International Conference on Learning Representations (ICLR)*, 2022. URL https://arxiv.org/abs/2110.06207.

Vikas Verma, Alex Lamb, Christopher Beckham, Amir Najafi, Ioannis Mitliagkas, David Lopez-Paz, and Yoshua Bengio. Manifold mixup: Better representations by interpolating hidden states. In *International Conference on Machine Learning (ICML)*, 2019. URL https://arxiv.org/abs/1806.05236.

Da-Wei Zhou, Han-Jia Ye, and De-Chuan Zhan. Learning placeholders for open-set recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. URL https://arxiv.org/abs/2103.15086.

A Additional Figures

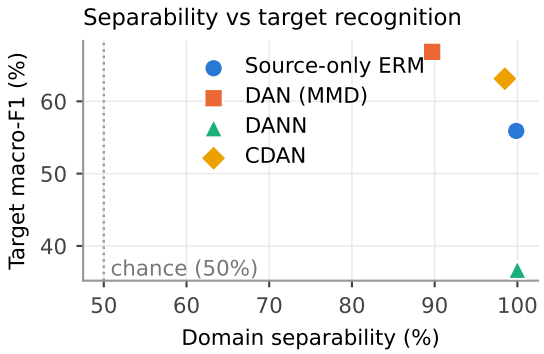


Figure 10: Target macro-F1 against source-vs-target separability (dotted: chance).

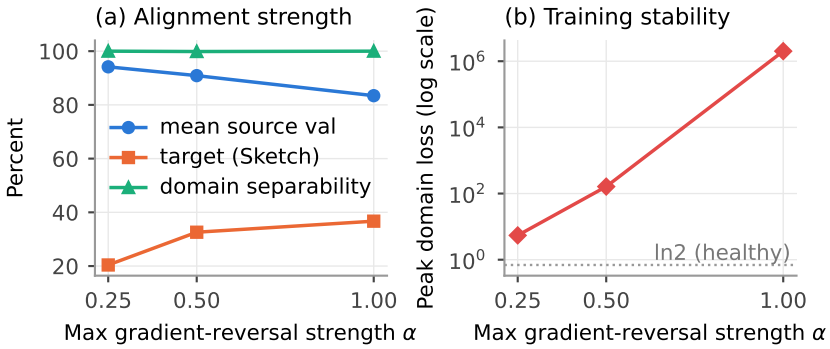


Figure 11: DANN $\alpha_{\max}$ study: source and target macro-F1, separability, and peak domain loss.

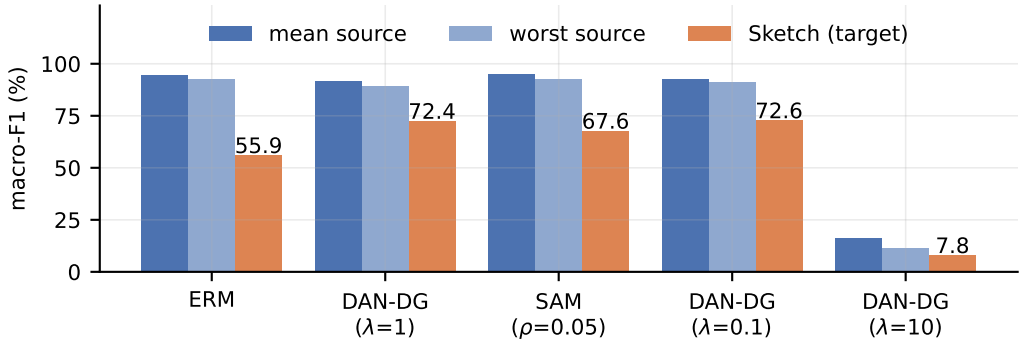


Figure 12: Mean-source, worst-source and Sketch macro-F1.

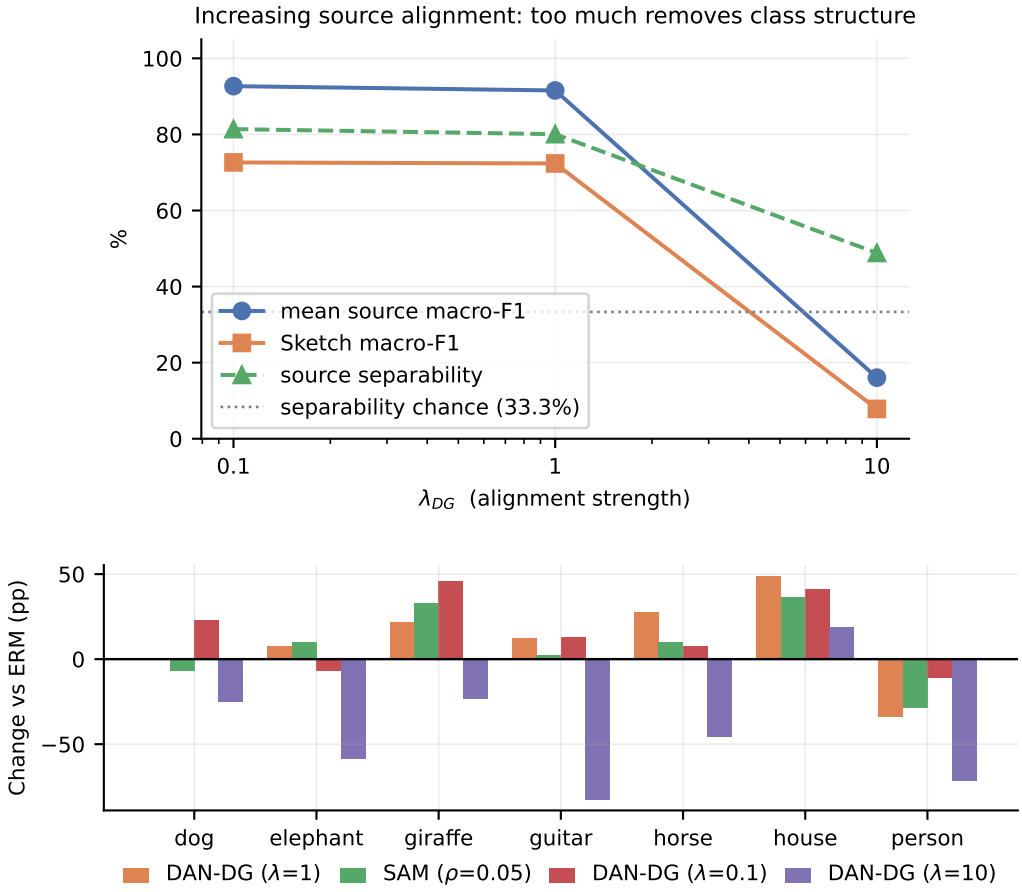


Figure 13: Top: λ_{DG} study. Bottom: per-class Sketch change against ERM.

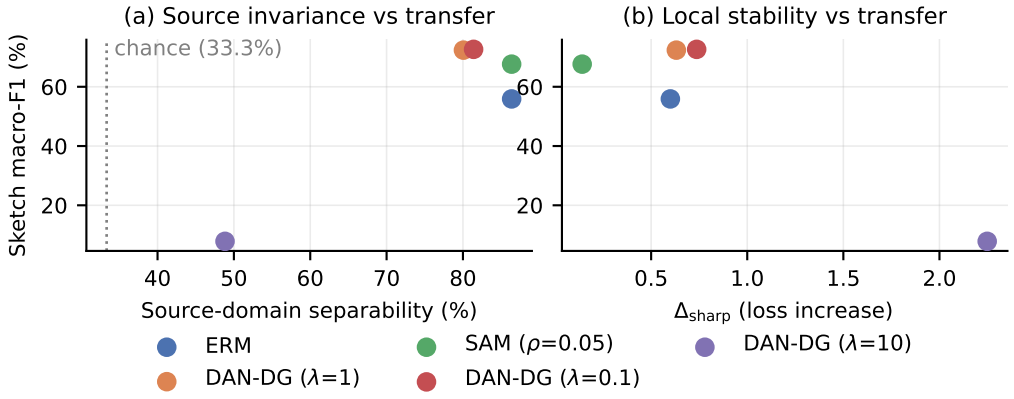


Figure 14: Sketch macro-F1 against (a) source separability and (b) Δ_{sharp}.

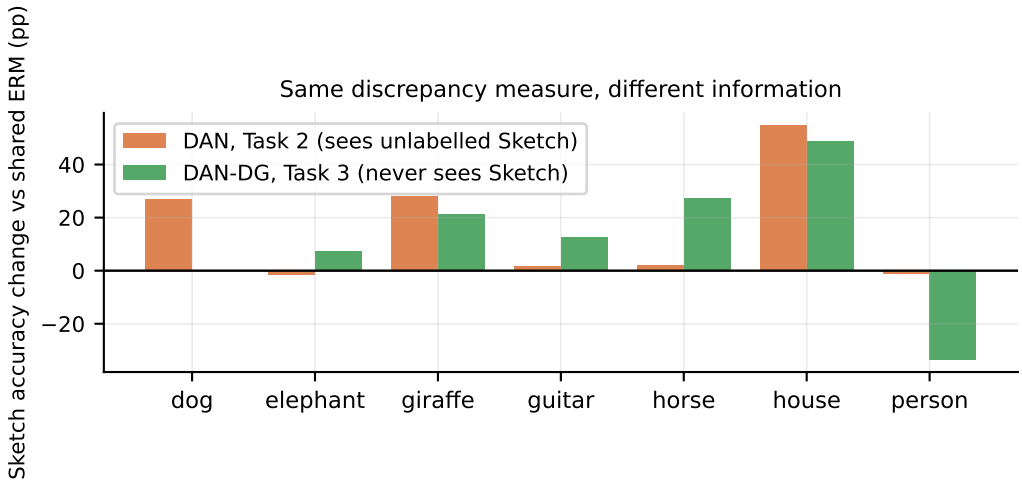


Figure 15: Per-class Sketch change against ERM for DAN (Task 2) and DAN-DG (Task 3).

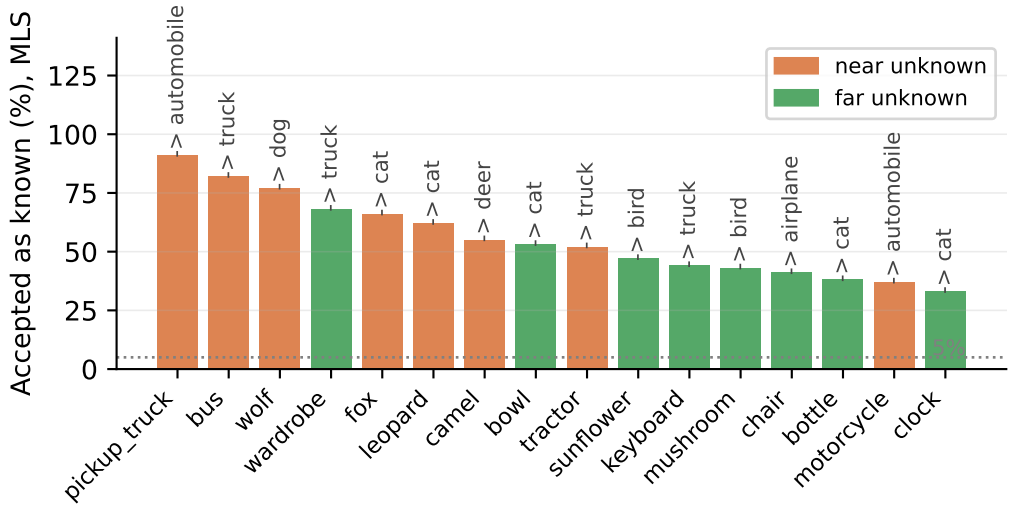


Figure 16: Acceptance per unknown class (Vanilla, MLS) with the CIFAR-10 class that absorbs most accepted images.

Table 10: Task 4 unknownness scores; larger $u(x)$ means more novel. z_k: logits, p_k: softmax probabilities, $K = 10$; $f(x)$: 512-d penultimate feature, μ_c: class means, Σ: shared diagonal covariance from CIFAR-10 training features with $\epsilon = 10^{-6}$ added to each diagonal entry.

Score	Unknownness	Captures
MSP	$u(x) = 1 - \max_k p_k(x)$	normalised confidence
MLS	$u(x) = -\max_k z_k(x)$	largest raw logit
Energy	$u(x) = -\log \sum_{k=1}^{K} e^{z_k(x)}$	all logits
Mahalanobis	$u(x) = \min_c (f(x) - \mu_c)^\top \Sigma^{-1} (f(x) - \mu_c)$	distance to nearest class

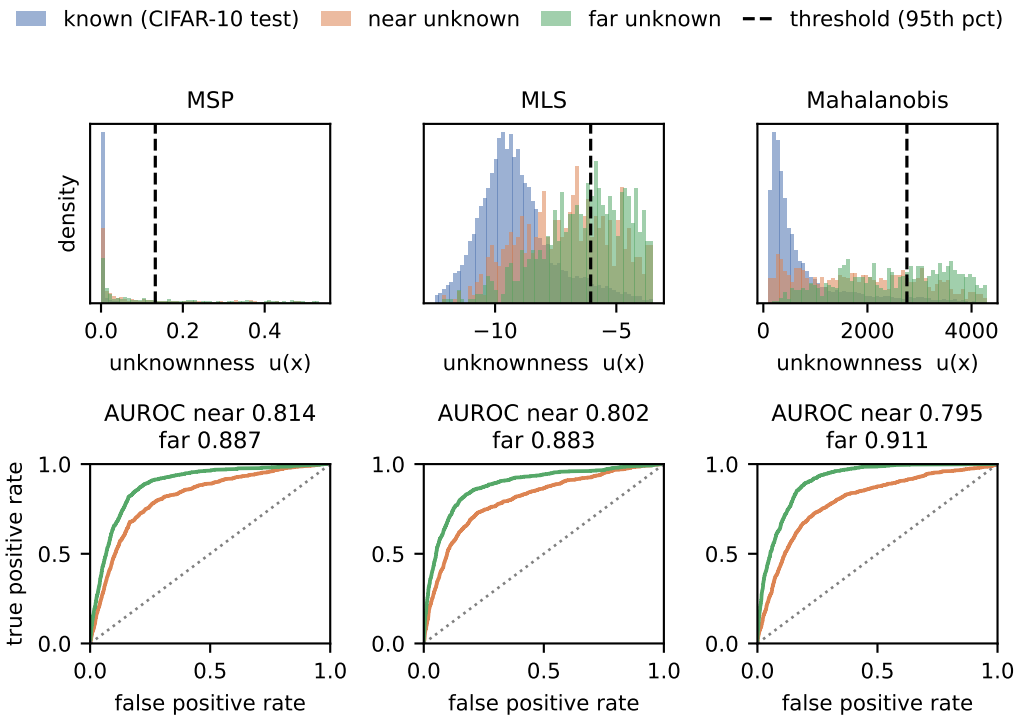


Figure 17: Unknownness distributions (top) and ROC curves with AUROC (bottom) for MSP, MLS and Mahalanobis on Vanilla.

B Experimental Configuration

Every value below is taken from the task configuration files (`task*/config.py`) and the saved run metadata. All experiments use seed 6304.

Table 11: Task 1: inductive biases and feature representations.

Setting	Configuration
Data	STL-10 test set; class-balanced evaluation subset of 500 images (50 per class); 224×224 inputs.
Backbones (frozen)	ResNet-50 (ImageNet-1k, `IMAGENET1K_V2`), ViT-B/16 (ImageNet-1k), CLIP ViT-B/32.
Decision methods	Linear head on each frozen backbone; CLIP zero-shot with prompt "a photo of a {}.".
Head training	AdamW, lr 10^{-3}, weight decay 10^{-4}, batch 64, at most 50 epochs, early stopping patience 5, 20% validation split.
Colour interventions	Grayscale; hue rotation by 30°.
Spatial interventions	Patch shuffle: 4×4 grid of 56×56 blocks, one fixed non-identity permutation per image. Translation: 8, 16, 32 px in four directions.
Cue conflict	AdaIN style transfer, $\alpha = 0.7$; five class pairs (airplane/bird, bird/car, monkey/truck, car/deer, bird/cat), both directions; 200 images retained after visual review.
Metrics	Top-1 accuracy, prediction consistency, shape bias, coverage, texture rank, cosine stability I_T.
Statistics	Paired McNemar tests with Holm correction; bootstrap intervals (shape bias), Wilson intervals (coverage).
Representation plot	L_2-normalised features, PCA, then joint t-SNE per backbone (perplexity 30).

Table 12: Task 2: unsupervised domain adaptation on PACS.

Setting	Configuration
Data	PACS, 7 classes. Sources (labelled): Photo, Art Painting, Cartoon, each split 80/20 stratified into train/validation. Target: Sketch, unlabelled during training, evaluated once in full.
Model	ResNet-18 (ImageNet, `IMAGENET1K_V1`) with a seven-class head, fine-tuned end to end; resize 256, crop 224.
Optimisation	AdamW, lr 10^{-4}, weight decay 10^{-4}, gradient clipping at norm 1.0, at most 30 epochs, early stopping patience 5.
Batches	8 images per source domain (24 source) plus 24 target images.
Model selection	Mean source-validation macro-F1; no target labels.
DAN	MMD weight $\lambda = 1$; three RBF kernels with bandwidths $0.5, 1, 2\times$ the median pairwise squared distance.
DANN	Domain discriminator (hidden 256, dropout 0.5), domain loss weight 1, gradient reversal up to $\alpha_{\max} = 1$.
CDAN	Class-conditioned domain discriminator with the same settings.
Controlled study	DANN $\alpha_{\max} \in \{0.25, 0.5, 1\}$, all other settings fixed.
Separability probe	Balanced logistic regression, $C = 1$, 70/30 split; source vs. target (chance 50%).

Table 13: Task 3: domain generalization to Sketch.

Setting	Configuration
Shared protocol	Identical to Task 2: splits, ResNet-18, optimiser, epoch budget, early stopping, 8 images per source domain per batch. Sketch is loaded only by the final evaluation.
ERM	The Task 2 Source-only checkpoint, reused unchanged.
DAN-DG	Classification loss plus the average MMD^2 over the three source pairs on the 512-d feature, same kernels as Task 2; $\lambda_{\mathrm{DG}} = 1$.
SAM	Non-adaptive SAM, $\rho = 0.05$, on AdamW with the ERM learning rate and weight decay.
Model selection	Mean source-validation macro-F1.
Controlled study	$\lambda_{\mathrm{DG}} \in \{0.1, 1, 10\}$, all other settings fixed.
Separability probe	Multinomial logistic regression, $C = 1$, 70/30 split over the three source domains (chance 33.3%).
Sharpness proxy	Loss increase after one normalised gradient-ascent step of radius 0.05 on a fixed validation batch of 32 images per source domain.

Table 14: Task 4: open-set recognition on CIFAR-10.

Setting	Configuration
Known data	CIFAR-10, ten classes; training partition split 90/10 stratified into train/validation; 32×32 inputs.
Unknown data	CIFAR-100, evaluation only, 100 images per class. Near (800): bus, pickup truck, motorcycle, tractor, wolf, fox, leopard, camel. Far (800): bottle, bowl, chair, clock, keyboard, mushroom, sunflower, wardrobe.
Model	CIFAR ResNet-18: 3×3 stride-1 first convolution, no initial max-pooling.
Vanilla training	SGD, lr 0.1, momentum 0.9, weight decay 5×10^{-4}, cosine decay, batch 128, 100 epochs; random crop (padding 4) and horizontal flip; checkpoint with the best validation accuracy.
GCSC	Vanilla recipe plus RandAugment (2 operations, magnitude 9).
PROSER	Fine-tuned from the Vanilla checkpoint for 50 epochs, lr 10^{-3}; 5 dummy classifiers; classifier-placeholder weight $\beta = 1$, data-placeholder weight $\gamma = 0.1$; manifold mixup after layer 2 with $\lambda \sim \mathrm{Beta}(2, 2)$.
Scores	MSP, MLS, Energy and Mahalanobis (shared diagonal covariance, $\epsilon = 10^{-6}$); see Table 10.
Rejection threshold	95th percentile of unknownness on CIFAR-10 validation data.
Metrics	CSA, AUROC (near, far, all), rejection rates, FPR@95TPR.